

PALLETPILOT · DESIGN DECISION RECORD

PalletPilot TRL 2 review decisions

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DOCUMENT CONTROL

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Status	Draft	Date	2026-09-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/palletpilot		
Source	docs/decisions/0001-trl2-review-decisions.md @ f9bb74f		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record Amish's decisions on the TRL 2 review items and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted for items D1 to D9; items O4 to O6 decided on 2026-09-25 in PLP-DDR-002; items O1 to O3 remain proposed

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed nine items as "Proposed, awaiting Amish", each with a recommendation. On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." For PalletPilot the instruction added that the approved safety recommendation is the low-cost lidar or time-of-flight stop layer (option 1A), that the approved budget is about \$1,550, and that the pitch changes to "layered stopping". Every item with a recommendation is therefore decided in favor of it. Items without a recommendation stay open.

The same instruction approved portfolio-wide SwapCell interface changes (a wake method for hosts without CAN, a charge-while-discharging mode, a latch vibration rating) and pricing shared SwapCell packs once. They do not apply to PalletPilot, which uses its own 24 V pack (D4). It also confirmed that co-design partners are picked per area later.

■ Options considered

The options for each item are those in docs/REVIEW.md (TRL 2 session) and in PLP-PRC-001 v0.2, Key design choices.

■ Decision

Table 1. Decided items.

#	Item	Decision
D1	Follow-mode stopping and the pitch	Option A: a low-cost 2D lidar or time-of-flight sensor is the main stopping layer in follow mode, the contact bumper is the last layer, follow mode runs at 0.6 m/s or less, and trials are for research in a closed area. A safety-rated laser scanner (option B) is named as the route to workplace use. The pitch changes from "bumper-based stopping" to "layered stopping". Decided by Amish, 2026-09-25: go with recommendation.
D2	Budget	Raise <code>budget_usd</code> from \$1,200 to \$1,550 (the figure given with option 1A). The donor jack stays excluded. Decided by Amish, 2026-09-25: go with recommendation.
D3	Drive layout	Two hub motors on a sprung module on the steering yoke, steering by differential drive in follow mode. Decided by Amish, 2026-09-25: go with recommendation.
D4	Power system	24 V (25.6 V nominal) LiFePO4, 20 Ah, not a SwapCell pack. Decided by Amish, 2026-09-25: go with recommendation.
D5	Mounting	Pack and electronics on the steering yoke, not the forks, checked at TRL 3 for handle clearance and bearing load. Decided by Amish, 2026-09-25: go with recommendation.
D6	Speeds	Walk 1.2 m/s handle-end first, 0.8 m/s forks first, creep 0.3 m/s, follow 0.6 m/s. Follow mode stays at 0.2 m/s or less until the lidar layer is built and tested. Decided by Amish, 2026-09-25: go with recommendation.
D7	Kit rating	1,000 kg design load; 1,500 kg maximum at 0.8 m/s or less, below the donor's 2,500 kg. Decided by Amish, 2026-09-25: go with recommendation.
D8	Legal route for workplace use	Document the kit for research use now, and approach a jack maker for written approval under 29 CFR 1910.178(a)(4) before any workplace trial. Decided by Amish, 2026-09-25: go with recommendation.
D9	First site partner types	One small warehouse and one maker space. Decided by Amish, 2026-09-25: go with recommendation. The named partners remain open (O1).

Table 2. Items left open at v0.1 (no recommendation was made, or raised at TRL 3). O4 to O6 are now decided in PLP-DDR-002.

#	Item	Status
O1	Named site and co-design partners	Proposed, awaiting Amish; partners are picked per area later
O2	Classification of follow mode under ISO 3691-4 and the safety functions it then needs	Proposed, awaiting Amish; no recommendation made
O3	Which donor jack models and steering yokes to support first	Proposed, awaiting Amish; no recommendation made
O4	Bumper-only speed or bumper travel (R8): 0.15 m/s limit or an edge with 65 mm or more of travel	Decided by Amish, 2026-09-25: go with recommendation (0.15 m/s limit); see PLP-DDR-002
O5	Closing the mass (1.7 kg) and cost (\$30) overruns, and the R5 bearing gap	Decided by Amish, 2026-09-25: go with recommendation (accept for now, recheck at the motor quote, no budget change; R5 fixes evaluated at TRL 4, on hold); see PLP-DDR-002
O6	A second contactor or rated safe torque off for PL d (R9)	Decided by Amish, 2026-09-25: go with recommendation (second contactor); see PLP-DDR-002

■ Consequences

- `project.yaml`: `budget_usd` is 1550 and the pitch reads "layered stopping". The problem line is unchanged, since no rewording was recommended.
- PLP-PRB-001, PLP-PRC-001 and PLP-REQ-001 are revised to v0.3. R7 is redefined as the lidar layer's stop, R17 carries the \$1,550 target, R2 and R8 carry the decided rating and speeds, and the design choices are no longer "proposed".
- The model, drawing PLP-DWG-001, calculation note PLP-CAL-001 and the BOM (new line 18, obstacle lidar) use the decided configuration.
- Workplace use stays excluded until D8's approval route and O2 are settled.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building or testing.